

Analysis of the polynomial $x^3 + 3x + 1$

In this worksheet we will study the field $K = \mathbb{Q}[\alpha] = \mathbb{Q}[X]/\langle P(X) \rangle$ where $P(X) = X^3 + 3X + 1$.

Routine algebraic calculations will be carried out using the Python package `sympy`.

```
# from sympy import var, discriminant, resultant,\
# factor, factorint, expand, div
from sympy import *
```

```
var('x')
pol = x**3 + 3*x + 1
pol
```

$$x^3 + 3x + 1$$

For various calculations we will need the derivative of $P(X)$ with respect to X .

```
dpol=pol.diff(x)
dpol
```

$$3x^2 + 3$$

In order to study the ring $\mathbb{Z}[\alpha]$ we will need the discriminant of $P(X)$ with respect to X .

```
discriminant(pol,x)
```

$$-135$$

Since the discriminant is negative, there is one real root and two complex roots.

We find the prime factors of the discriminant to know at what primes p we *could* the ring of integers to be *larger* than $\mathbb{Z}_{(p)}[\alpha]$.

```
factorint(135)
```

$$\{3: 3, 5: 1\}$$

Thus, we have $(-135) = (-1) \cdot 3^3 \cdot 5$.

Analysis for $p = 5$

Since $P(X)$ has a multiple root modulo $p = 5$, it has a common root with $P'(X)$ in $\mathbb{F}_5$. We wish to find this root.

To do so, we use the GCD algorithm to calculate a common factor of $P(X)$ and $P'(X)$ in $\mathbb{F}_5$.

To use the GCD algorithm we first replace $P'(X)$ by a multiple that is monic in $\mathbb{F}_5$.

Thus, in $\mathbb{F}_5$ we can use $P_1(X) = X^2 + 1$ in place of $P'(X)$.

```
dpol1=x**2+1
```

The remainder on division of $P(X)$ by $P_1(X)$ *must* be the common linear factor.

```
lin=div(pol,dpol1)[1]
lin=(lin.coeff(x)%5)*x+lin.coeff(x,n=0)%5
lin
```

$2x + 1$

Make this monic in $\mathbb{F}_5$

```
lin=3*lin
lin=(lin.coeff(x,n=1)%5)*x+lin.subs({x:0})%5
lin
```

$x + 3$

We now find the remaining linear factor of $P(X)$ over $\mathbb{F}_5$.

```
olin=div(pol,lin**2)[0]
b=pow(olin.coeff(x,n=1),5-2,5)
olin=b*olin
olin=(olin.coeff(x,n=1)%5)*x+olin.subs({x:0})%5
olin
```

$x + 4$

This gives us the two roots of $P(X)$ in $\mathbb{F}_5$.

```
(5-olin.coeff(x,n=0), 5-lin.coeff(x,n=0))
(1, 2)
```

Here 1 is a simple root and 2 is a double root.

Just a quick check that $(X - 2)^2 \cdot (X - 1)$ equals $P(X)$ modulo 5.

```
check=expand(pol-(x-2)**2*(x-1))
[check.coeff(x**i)%5 for i in [0,1,2]]
[0, 0, 0]
```

Since 1 is a simple root, we can use Hensel lifting to lift it to a root η in $\mathbb{Z}_5$.

Thus, $P(X)$ becomes $(X - \eta)Q(X)$ in $\mathbb{Z}_5[X]$ for some quadratic polynomial $Q(X)$ in $\mathbb{Z}_5[X]$.

We will prove that the image of $Q(X)$ is irreducible in $\mathbb{Z}[X]/\langle 5^2 \rangle$. It will follow that $Q(X)$ is irreducible in $\mathbb{Z}_5[X]$.

Now $Q(X) = (X - 2)^2$ in $\mathbb{F}_5[X]$. Thus, the ring $R = \mathbb{Z}_5[X]/\langle Q(X) \rangle$ is a ramified extension of degree 2 of $\mathbb{Z}_5$.

It follows that

$$\frac{\mathbb{Z}_5[X]}{\langle P(X) \rangle} = \mathbb{Z}_5 \times R$$

Hensel lifting the simple root

We first calculate η modulo 5^2 .

To do this we need to calculate the inverse of $P'(1)$ in $\mathbb{F}_5$.

```
dpol.subs({x:1})%5
```

```
1
```

Thus, we can calculate the Hensel lift.

```
rt2=(1-pol.subs({x:1}))%(5*5)
rt2
```

```
21
```

Check that this is a root!

```
pol.subs({x:rt2})%(5*5)
```

```
0
```

We divide $P(X)$ by $(X - 21)$ to get the quadratic polynomial $Q(X)$ modulo 5^2 .

```
qpol=div(pol,x-rt2)[0]
qpol
```

```
 $x^2 + 21x + 444$ 
```

We first note that $Q(X) - (X - 2)^2$ is divisible by 5.

```
rqp1=expand(qpol-(x-2)**2)
rqp1.coef(x,n=1)%5,rqp1.coef(x,n=0)%5
```

```
(0, 0)
```

We then use the Eisenstein criterion to prove that $Q(X)$ is irreducible in $\mathbb{Z}[X]/\langle 5^2 \rangle$.

```
sqpol=expand(qpol.subs({x:x+2}))
sqpol.coef(x,n=1)%5,sqpol.coef(x,n=0)%(5**2)
```

```
(0, 15)
```

Since $Q(X)$ in $\mathbb{Z}_5[X]$ is a lift of this polynomial, it is also irreducible.

As seen above this means that $\mathbb{Z}_5[\alpha]$ is of the form $\mathbb{Z}_5 \times R$ where R is a ramified extension of degree 2 of $\mathbb{Z}_5$.

It follows that $\mathbb{Z}_5[\alpha]$ is the ring of integers $\mathbb{Z}_5 \otimes_{\mathbb{Z}} \mathcal{O}_K$.

Analysis at $p = 3$

We note that modulo $p = 3$, the polynomial $P(X)$ becomes $X^3 + 1 = (X + 1)^3$.

Thus, a substitution of X by $X - 1$ will give us a polynomial $C(X)$ whose reduction in $\mathbb{F}_3[X]$ is X^3 .

We use Eisenstein's criterion to see that $C(X)$ is irreducible. It then follows that $\mathbb{Z}_3[\alpha] = \mathbb{Z}_3[X]/\langle P(X) \rangle$ is exactly the ring of integers in $\mathbb{Z}_3 \otimes_{\mathbb{Z}} \mathcal{O}_K$.

```
cupol=expand(pol.subs({x:x-1}))
cupol

$$x^3 - 3x^2 + 6x - 3$$

```

Other primes

At primes not dividing the discriminant, the extension of $\mathbb{Z}_p$ is unramified. In this case, the nature of the splitting of $\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathcal{O}_K$ is completely determined by the splitting of $\mathbb{F}_p[\alpha] = \mathbb{F}_p[X]/\langle P(X) \rangle$.

We have distinct possibilities based on whether the discriminant is a square or not.

1. If the discriminant is a square modulo p , then the extension of $\mathbb{F}_p$ where a root lies is either $\mathbb{F}_p$ itself or a cyclic extension.
2. If the discriminant is *not* a square modulo p , then the polynomial has one root in $\mathbb{F}_p$ and the other two roots in a quadratic extension of $\mathbb{F}_p$ (where the square root of the discriminant lies).

Thus, we can examine different primes depending on the behaviour of -135 modulo p .

Case p is 2

In this case we see that $P(X) = X^3 + 3X + 1$ has no roots in $\mathbb{F}_2$. Hence, all the roots lie in $\mathbb{F}_{2^3}$.

Case p is 7

In this case we see that -135 is not a square modulo 7.

```
pow(-135, (7-1)//2, 7)
```

6

So we look for a root in $\mathbb{F}_7$.

```
cpol=div(x**6-1, pol)[1]
cpol

$$9x^2 + 6x$$

```

```

cpol=4*cpol
cpol=(cpol.coeff(x,2)%7)*x**2 + (cpol.coeff(x,1)%7)*x \
      + cpol.coeff(x,0)%7
cpol
 $x^2 + 3x$ 
lin=div(pol,cpol)[1]
lin
 $12x + 1$ 
lin=3*lin
lin=(lin.coeff(x,1)%7)*x + lin.coeff(x,0)%7
lin
 $x + 3$ 

```

We check that -3 is a root of $P(X)$ in $\mathbb{F}_7$.

```

pol.subs(x,-3)%7
0

```

We check that $Q(X) = P(X)/(X + 3)$ is irreducible in $\mathbb{F}_7[X]$.

```

qpol=div(pol,x+3)[0]
qpol
 $x^2 - 3x + 12$ 
pow(discriminant(qpol),(7-1)//2,7)
6

```

Thus, $\mathbb{F}_7[X]/\langle Q(X) \rangle$ is a quadratic extension of $\mathbb{F}_7$.

Find p

We wish to look for p such that -135 is a square modulo p and there is a root of $X^3 + 3X + 1$ in $\mathbb{F}_p$. In this case, *all* roots of the equation are in $\mathbb{F}_p$.

To satisfy the first condition, we need:

- -15 is a square modulo p .

To satisfy the second condition, we need:

- There is an integer n such that p divides $P(n)$.

```

[pol.subs(x,i) for i in range(-5,6)]
[-139, -75, -35, -13, -3, 1, 5, 15, 37, 77, 141]

```

Thus, candidates are $p = 139, 13, 37, 11, 47$.

We check which of these p is such that -15 is a square modulo p

```

for p in [139,13,37,11,47]:
    if pow(-15,(p-1)//2,p)==1:
        print(p)

```

```

139
47

```

The smallest prime in this list is $p = 47$.

We find the roots of $P(X)$ in $\mathbb{F}_{47}$.

```

for i in range(47):
    val=pol.subs({x:i})%47
    if val==0:
        print(i)

```

```

5
6
36

```

It follows that $P(X)$ splits into linear factors in $\mathbb{F}_{47}$.